DO NOW — RECALL

Lesson 075 · Big O: Quadratic, Exponential and Growth

Name: _____ Class: _____ Date: _____

1. How does O(n) differ from O(1)?

2. How many halvings 32→1?

3. Which algorithm naturally explores include/exclude branches?

4. A rectangular 4×3 zero-indexed array holds how many elements? Give its last valid index pair.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

One grows with n; the other does not.

2. Model answer

Five.

3. Model answer

Subset backtracking or exhaustive enumeration.

4. Model answer

12 elements; last index (3,2).

By the end, you can…

• Two nested n loops perform n×n operations, producing $O(n^2)$ time when each inner operation is constant.

• A binary decision at each of n positions can create 2^n possibilities. This exponential growth quickly outruns polynomial methods.

• Big O ignores constants and lower-order terms; compare classes by input growth while acknowledging memory, workload and implementation detail.

Vocabulary — know these terms

quadratic — Growth proportional to n^2.

exponential — Growth like 2^n.

pruning — Avoid exploring impossible branches.

scalability — Handling larger inputs feasibly.

NEW LEARNING

Read, annotate and explain

Observe

Two nested n loops perform n×n operations, producing $O(n^2)$ time when each inner operation is constant.

Explain

A binary decision at each of n positions can create 2^n possibilities. This exponential growth quickly outruns polynomial methods.

Evaluate

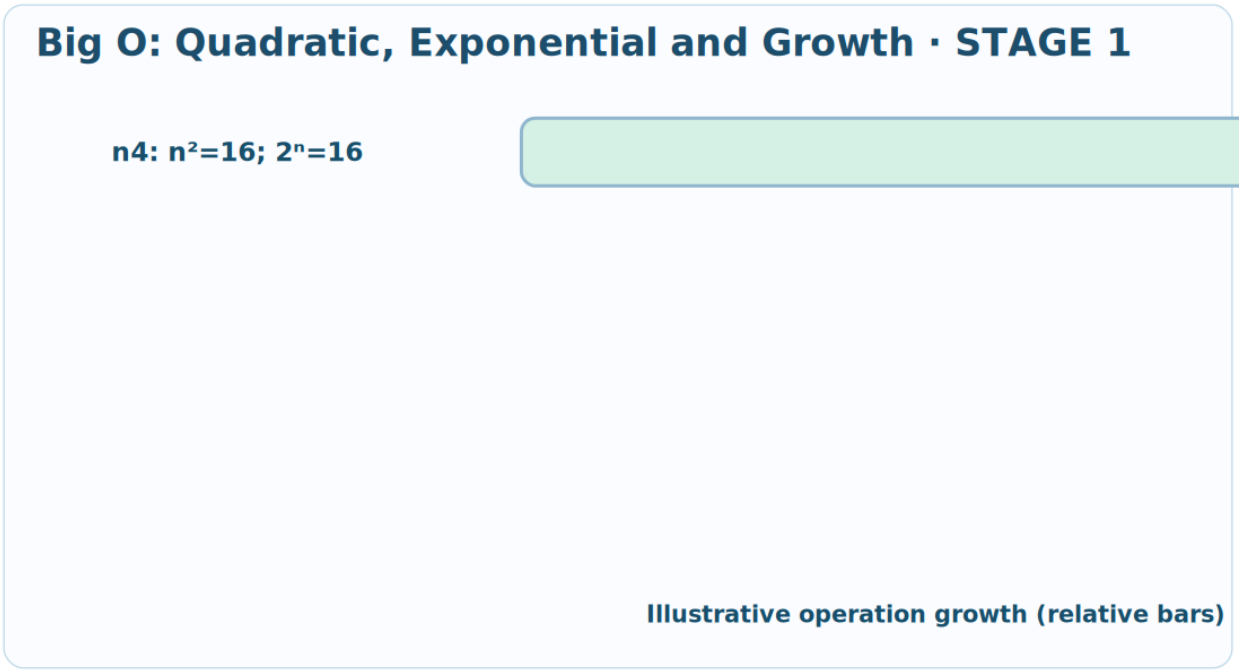
Big O ignores constants and lower-order terms; compare classes by input growth while acknowledging memory, workload and implementation detail.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning

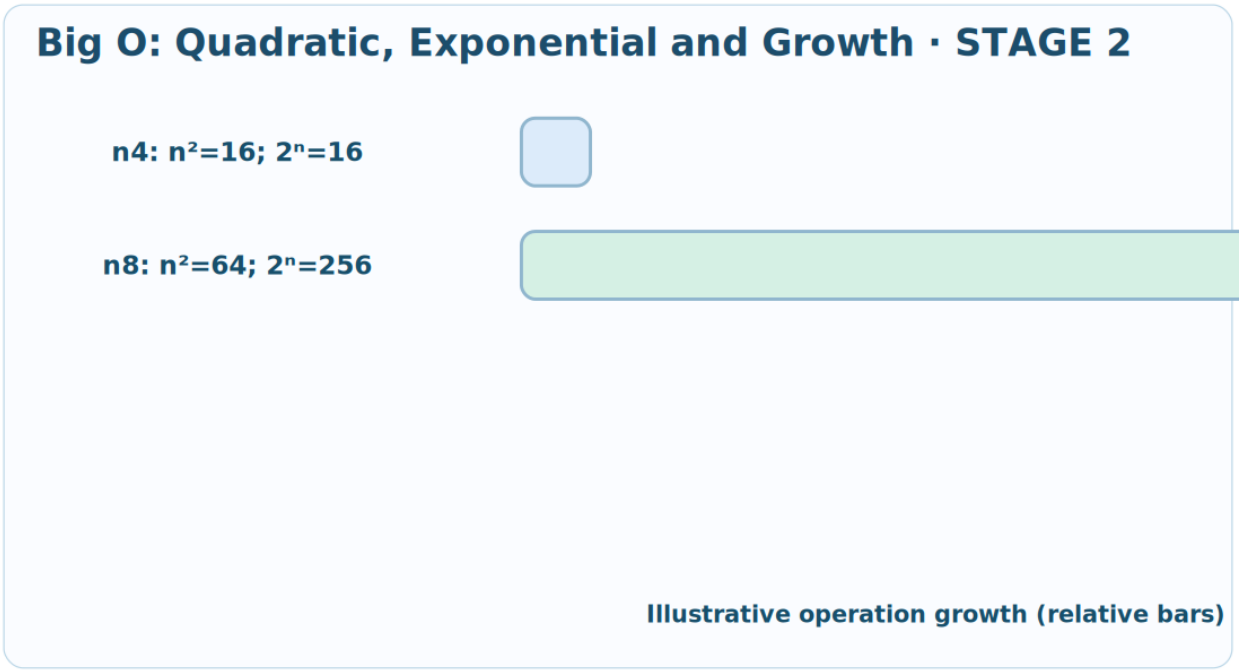


Big O: Quadratic, Exponential and Growth · STAGE 1: n4: $n^2=16$; $2^n=16$

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning

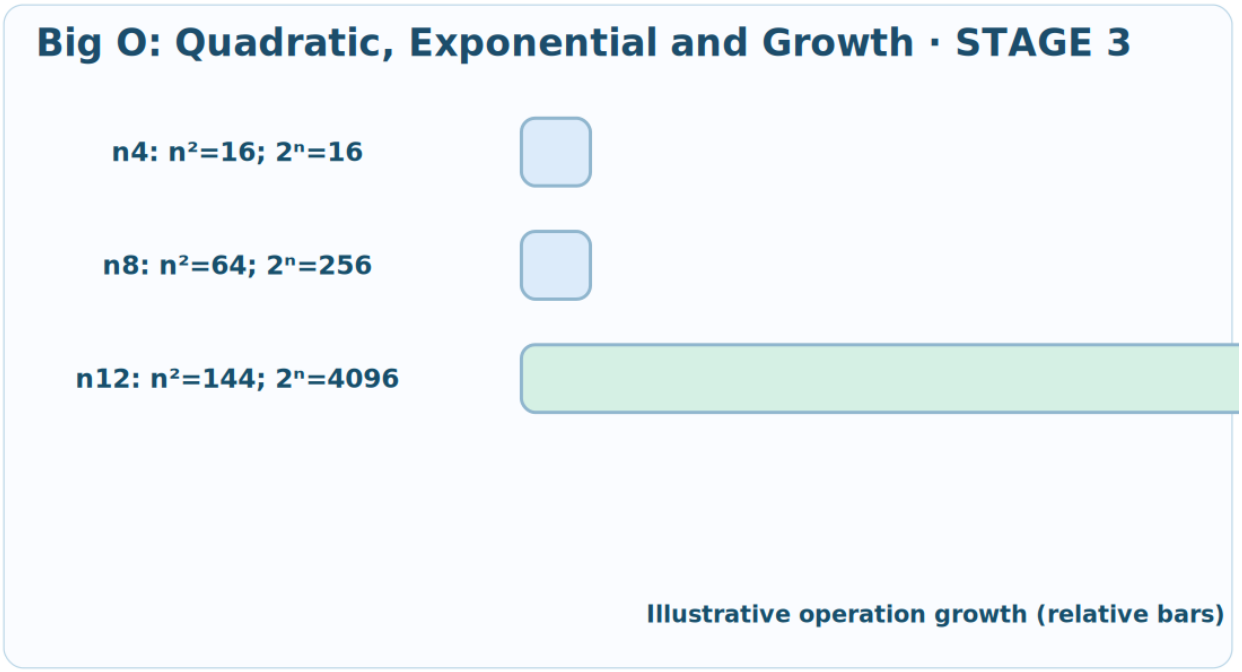


Big O: Quadratic, Exponential and Growth · STAGE 2: n4: n²=16; 2ⁿ=16; n8: n²=64; 2ⁿ=256

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Big O: Quadratic, Exponential and Growth · STAGE 3: n4: n²=16; 2ⁿ=16; n8: n²=64; 2ⁿ=256; n12: n²=144; 2ⁿ=4096

I DO — WORKED EXAMPLE

Follow the worked method

For n=6, calculate total iterations of two independent nested n loops.

Explain

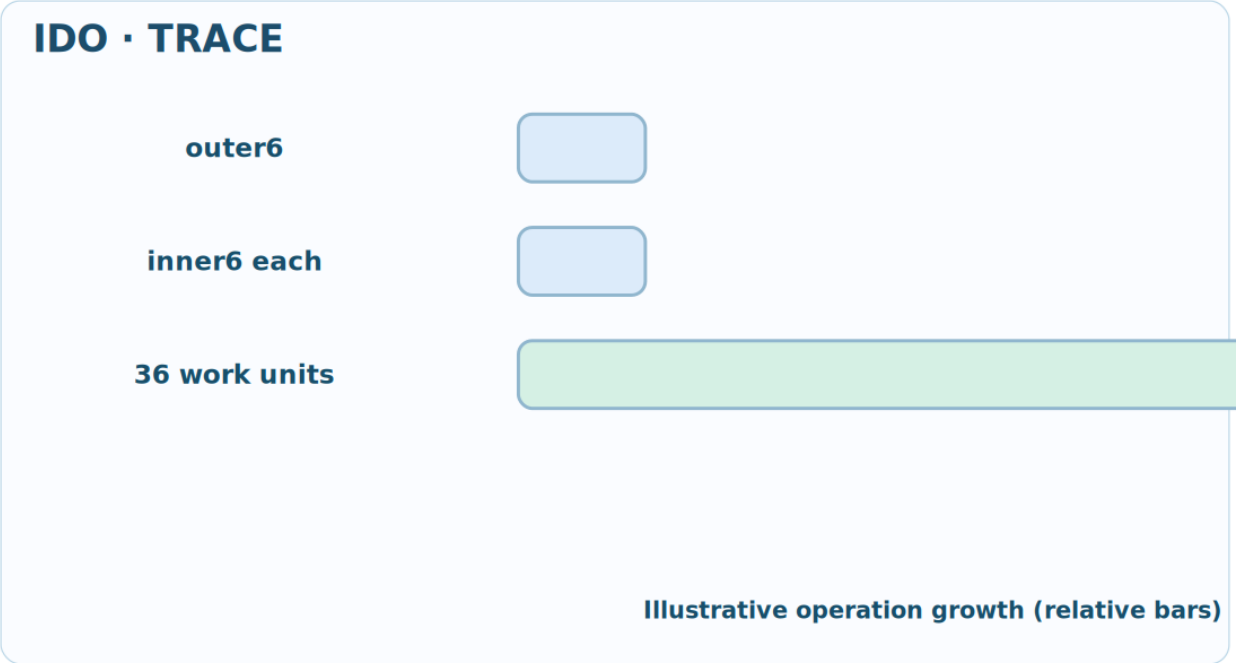
Each of the six outer iterations runs six inner iterations.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

For n=6, calculate total iterations of two independent nested n loops.

Each of the six outer iterations runs six inner iterations.



IDO · TRACE: outer6; inner6 each; 36 work units

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

TRY: Attempt the guided questions in BLACK PEN. DISCUSS: Compare approaches with a partner, explain your reasoning and agree one improvement. IMPROVE: After the Exit Ticket, use the source CHECK model in GREEN PEN to correct and explain errors.

Guided 1 — TRY

For n=6 decisions, calculate binary include/exclude combinations.
Source code / input:
2^6=64 possibilities, O(2^n) in full enumeration.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations or a justified explanation.

Question 1

For n=5 and n=10, calculate n^2 and 2^n values. [6]

Question 2

Explain time growth of a nested n×n loop and classify. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations or a justified explanation.

Question 3

A problem has 30 binary choices. Evaluate full subset enumeration and one way to reduce exploration. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking.

An $O(2^n)$ approach grows only twice as much when n doubles.

Circle: Confident / Getting there / Need help

Explain what you can now do independently.

GREEN PEN — Correct and explain

Choose one error. Write the correct method and explain precisely why your first response was wrong.

WE DO — CHECK AND IMPROVE

Source-grounded model answers

CHECK only after completing the guided practice and Exit Ticket. Use GREEN PEN to improve your method and explain any error.

Guided 1 — CHECK

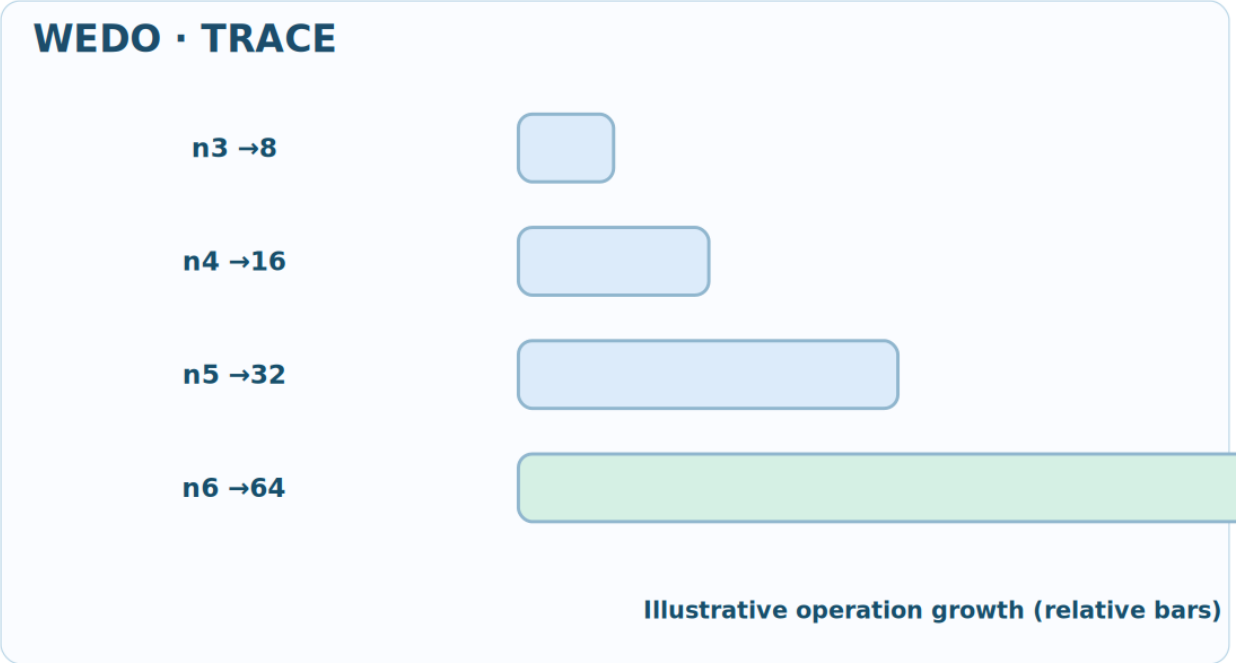
Explain
Every new independent binary choice doubles the count.

We Do source CHECK · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

For n=6 decisions, calculate binary include/exclude combinations.

Every new independent binary choice doubles the count.



WEDO · TRACE: n3 →8; n4 →16; n5 →32; n6 →64